

$$\begin{aligned}
 \Phi &= \iint_S F(x,y,z) \cdot n(x,y,z) \, dS \\
 &= \iint_S (P \cos \alpha + Q \cos \beta + R \cos \gamma) \, dS \\
 &= \iint_S P \, dy \, dz + Q \, dz \, dx + R \, dx \, dy
 \end{aligned}$$

例: 计算 $\iint_S xdydz + ydzdx + zdx dy$ S 为上半球面
 $x^2 + y^2 + z^2 - 1 = 0$ 的上侧

法一: (转化为第一类)

$$\iint_{S \text{ 上侧}} Pdydz + Qdzdx + Rdx dy = \iint_S (P \cos \alpha + Q \cos \beta + R \cos \gamma) ds$$

$$P=x \quad Q=y \quad R=z$$

$$S: x^2 + y^2 + z^2 - 1 = 0 \quad \vec{n} = (x, y, z)$$

$$\text{原式} = \iint_S (x^2 + y^2 + z^2) ds = \iint_S ds = 2\pi$$

法二: $P=x, Q=y, R=z$

$$S: z = \sqrt{1-x^2-y^2} \quad (x, y) \in D_{xy}$$

$$D_{xy}: x^2 + y^2 \leq 1$$

$$z'_x = -\frac{x}{\sqrt{1-x^2-y^2}} \quad z'_y = -\frac{y}{\sqrt{1-x^2-y^2}}$$

$$\text{原式} = \iint_{D_{xy}} \left(\frac{x^2}{\sqrt{1-x^2-y^2}} + \frac{y^2}{\sqrt{1-x^2-y^2}} + \sqrt{1-x^2-y^2} \right) dx dy$$

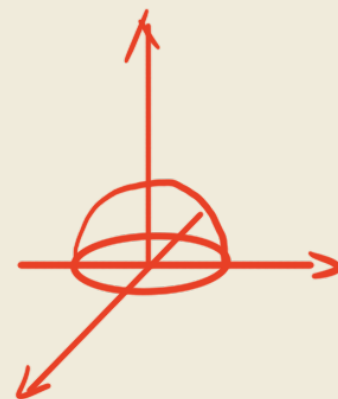
$$= \iint_{D_{xy}} \frac{1}{\sqrt{1-x^2-y^2}} dx dy$$

$$= \iint_{D_{xy}} \frac{\rho}{\sqrt{1-\rho^2}} d\rho d\theta = \int_0^{2\pi} d\theta \int_0^1 \frac{\rho}{\sqrt{1-\rho^2}} d\rho = 2\pi$$

$$\text{法三: } \iint_S z \, dx \, dy = \iint_{D_{xy}} \sqrt{1-x^2-y^2} \, dx \, dy = \int_0^{2\pi} d\theta \int_0^1 \rho \sqrt{1-\rho^2} \, d\rho = \frac{2\pi}{3}$$

$$\begin{aligned} \iint_S x \, dy \, dz &= \iint_{S_1 \text{ 前侧}} x \, dy \, dz + \iint_{S_2 \text{ 后侧}} x \, dy \, dz \\ &= \iint_{D_{yz}} \sqrt{1-y^2-z^2} \, dy \, dz - \iint_{D_{yz}} -\sqrt{1-y^2-z^2} \, dy \, dz = \frac{2\pi}{3} \end{aligned}$$

$$\begin{aligned} \iint_S y \, dz \, dx &= \iint_{S_3 \text{ 右侧}} y \, dz \, dx + \iint_{S_4 \text{ 左侧}} y \, dz \, dx \\ &= \iint_{D_{zx}} \sqrt{1-z^2-x^2} \, dz \, dx - \iint_{D_{zx}} -\sqrt{1-z^2-x^2} \, dz \, dx = \frac{2\pi}{3} \end{aligned}$$



例 计算 $\iint_S xy \, dy \, dz + z \, dx \, dy$ S 是圆柱面 $x^2 + y^2 = 1$ ($y \geq 0, 0 \leq z \leq 2$) 的右侧

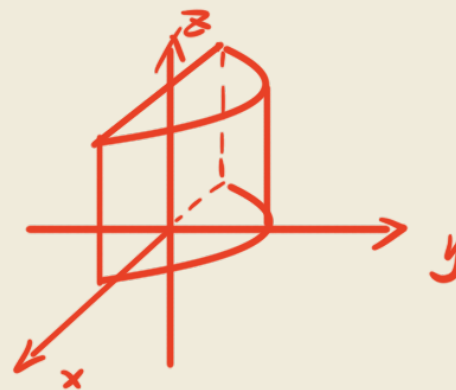
$$\text{法一: } S: y = \sqrt{1-x^2} \quad (z, x) \in D_{zx}$$

$$\iint_S P \, dy \, dz + Q \, dz \, dx + R \, dx \, dy = \iint_{D_{zx}} (P + y'_x) + Q + R(-y'_z) \, dz \, dx$$

$$\text{原式} = \iint_{D_{zx}} (x \cdot \sqrt{1-x^2} \cdot \frac{x}{\sqrt{1-x^2}} + z \cdot 0) \, dz \, dx$$

$$= \iint_{D_{zx}} x^2 \, dz \, dx$$

$$= \int_0^2 dz \int_{-1}^1 x^2 \, dx = \frac{4}{3}$$



法二: $\iint_S xy dy dz$

记 S 中 $x \geq 0$ 部分 $S_1: x = \sqrt{1-y^2}$ 前侧 $(y, z) \in D_{yz}$
 $x \leq 0$ 部分 $S_2: x = -\sqrt{1-y^2}$ 后侧

$D_{yz}: 0 \leq y \leq 1, 0 \leq z \leq 2$

$$\begin{aligned} \iint_S xy dy dz &= \iint_{S_1 \text{ 前侧}} xy dy dz + \iint_{S_2 \text{ 后侧}} xy dy dz \\ &= 2 \iint_{D_{yz}} y \sqrt{1-y^2} dy dz = 2 \int_0^2 dz \int_0^1 y \sqrt{1-y^2} dy = \frac{4}{3} \end{aligned}$$

$$\iint_S z dx dy = \iint_S z \cdot \cos r ds = 0$$

原式 = $\frac{4}{3}$